



MakersSoc Analogue Binary Counter

JJ Littleton

This workshop will guide members through building an analogue binary counter using dual D flip-flops. The counter will increment with each clock pulse, visually displaying the binary count using LEDs. This hands-on session will help develop a fundamental understanding of sequential logic circuits and flip-flop behaviour.

Equipment needed:

Wire

Dual D Flip Flop x2 – such as 74HC74

LEDs x4

220Ω resistors x5

Push button

Power supply (5V)

Breadboard

(Personal) Toolkit

Extension:

Solder

Soldering Iron

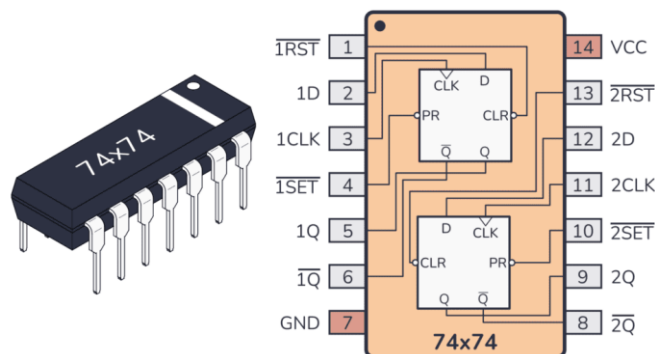
Soldering Iron safety/training

Strip board

Trace cutters

Inside the integrated chip & Understanding the Dual D Flip-Flop

[Show / Explain inside the analogue dual d flip flop]



A D flip-flop stores a single bit of data and changes its output state on the rising edge of a clock pulse. The 74HC74 contains two D flip-flops, each with a clock (CLK), data input (D), set (SET), reset (RST), and outputs (Q and Q⁻ (known as Q-bar)).

When the clock signal transitions from low to high:

If D = 1, Q is set to 1.

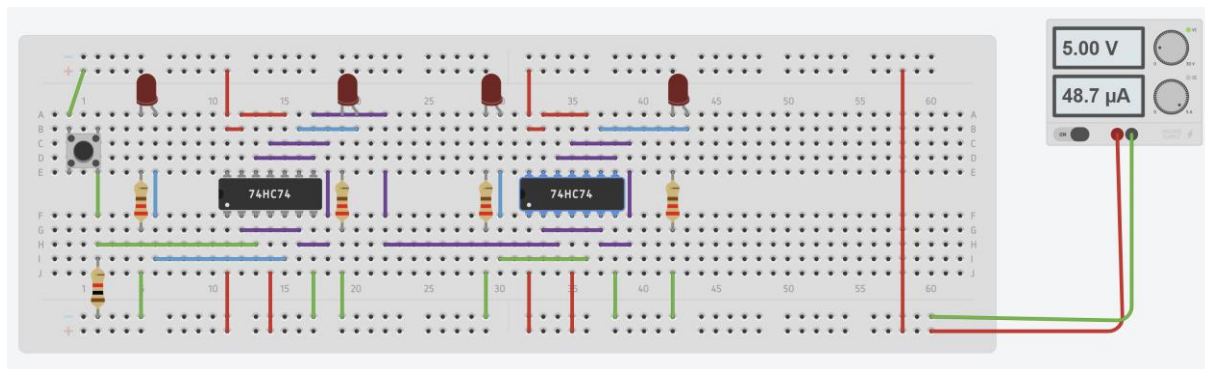
If D = 0, Q is set to 0.

By chaining two flip-flops together, we can create a 2-bit binary counter.



Circuit

1. Assemble this circuit. Do not connect to power supply until the very end.



We use the button press as a clock pulse for these chips.

The lab bread boards may be made of 2 smaller breadboards and power bridging across.

Power supply should be set to 5V.

Extension(s)

Try soldering it on strip board or extend the circuit with more flip-flops for a higher bit counter.

Replace the manual clock with a 555 timer to automate counting.

74x74 integrated chip image from:

<https://www.build-electronic-circuits.com/7400-series-integrated-circuits/74hc74-74ls74/>

Circuit modified from:

<https://www.tinkercad.com/things/hVI1Blp74UW-counter>